

L1-zkEVM breakout #05

May 2026



EIP-8025 has been proposed for Hegota!

- Proposed EIP-8025 (Optional proofs) for Hegota
- Brief summary of CL and EL side of the EIP.
- Argument special characteristic of an “optional” protocol feature
- [Youtube link](#) from ACDC (Francesco | Ignacio)
- [Forkcast Q&A section](#)

Agenda

- Intro
- Projects updates and Q&As.
- Update from Derek about `evm-asm`
- Update from Ben about input deserialization for zkVMs
- Update from Peter about BiB progress

zkEVM project 1 & 2

Execution Witness & Guest Program specs/test

May 2026



Specs & tests

- [zkevm@v0.4.0](#)
 - Add `schema-id` for future-proof compatibility of guest input bytes.
 - Enrich `ChainConfig` in stateless input to fully define the current active fork
 - Properly fill `public\_keys` to allow faster transaction signature verification.
- [zkevm@v0.4.1](#)
 - Add `engine-witness` simulator and `executionWitnessMutated` flag in fixtures.
 - Include `blockchain\_test\_engine` format as part of released fixtures.
- Upcoming in next release:
 - Fixture filling using external EL (i.e. Geth)
 - More tests with structurally invalid SSZ shape.
 - Update SSZ-max-lengths for execution witness types based on first principles.
 - New valid/invalid `ChainConfig` test cases.
 - New `public\_key` wrong parity important test case.

Hive-based fixture runner

- Both for:
 - Execution witness compliance
 - Stateless block validator (i.e. guest program) compliance
- Repo: <https://github.com/eth-act/eest-execution-witness-dashboard>
- Website: <https://eth-act.github.io/eest-execution-witness-dashboard/#/group/execution-witness>

The screenshot displays the Ethereum Hive Integration Testing dashboard. At the top, the header shows the Ethereum Hive logo and the text "Integration Testing". Below the header, the breadcrumb "Home / execution-witness" is visible. The main section is titled "execution-witness" and features a summary bar with a red square icon, a green circle with "0", a yellow triangle with "0", a red circle with "1", and a "1 Result" button. The last update is "Last: Jun 9, 2026 11:17".

Under the "Latest Test Results" section, there are filters for "Clients: All", "Test Suite: All", and a checkbox for "Hide old jobs (>10d)". Sorting options include "Sort by Name", "Sort by Coverage", "Sort by Time", "Group by Test", and "Group by Client".

A specific test result is highlighted for "eels/consume-engine-witness". It shows a sub-test "ethrex_rlp-engineapi" with a status of "Failing". The test results are: 18467 passes (green checkmark) and 1950 failures (red X). The timestamp is "Jun 9, 2026 11:17:20".

At the bottom, a table summarizes the test results:

DATE	TEST NAME	CLIENTS	TOTAL	PASS	FAIL	DIFF	STATUS
Jun 9, 2026 11:17:20	eels/consume-engine-witness	ethrex_rlp-engineapi	20417	18467	1950	—	Failing

Side explorations (security bugs found)

- [Found soundness bug in ArithEq/ArithEq384 in Zisk](#) [Ignacio]
- [Fix OOM critical bug in libssz](#) [Ignacio]

EL progress

- Ethrex:
 - [Ignacio] [Implement engine_newPayloadWithWitnessV5](#)
 - [Ben] [Resolve memory limit crash on zisk by removing redundant allocations](#)
 - [Ben] [Serialize execution witness with ssz instead of rkyv](#)
 - [Ben] [Route remaining hash calls through precompile](#)
 - [Ben] [Replace JUMP\(I\)-targets with bitvector](#)
- Nethermind:
 - [Team] Passes all the EEST stateless execution tests
 - [Team] Significantly improved performance, and are still boosting it alongside error handling
 - [Rubo | Ignacio] [Polish a bit more ChainConfig guest input design](#)
- Reth:
 - [Andrurachi] [Support rv64im zkVM target via lower-atomic pass](#)
- Geth:
 - [Ignacio] [Implement engine_newPayloadWithWitnessV5 + use spec ordering](#)
- Zilkworm:
 - [canepat] [Integrated Zilkworm into zkevm-benchmark-workload](#)
 - [developeruche] [Zisk Zilkworm guest program implementation without acceleration](#)
- Besu:
 - [Gabriel] Implement engine_newPayloadWithWitnessV5
 - [Gabriel] Zesu passing all latest tests – next steps working in CI pipeline to publish ELF.

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Benchmarking and Metrics



Updates

- [Ben] [Add optimized LLVM parameters for Zisk target](#)
- [Ignacio] [Add EEST adapter to newly evm-gasfit repo from Maria](#)
- Ethereum Foundation proving cluster is ready (❤️ ef devops & pandaops)